

Supplementary Material

Rule-Guided Weak Supervision for Entity-Aware Aviation Speech Recognition and Unified Text Restoration

Implementation and evaluation details. Matched adaptations use decoder-only rank-8 LoRA (scaling 16, dropout 0.05) on the decoder q_proj , k_proj , v_proj , and out_proj projections. AdamW uses learning rate 10^{-4} , micro-batch 1, accumulation 8, and 6,013 optimizer steps with seed 20260725 on one A800 GPU in half precision; the final-step checkpoint is evaluated. Matched LoRA and Auxiliary use unit ASR token weights, Entity Weighting uses 1.5 on aligned entity tokens, and Auxiliary uses $\lambda_{ent} = 0.2$ with $\omega(O) = 0.25$. Audio is resampled to 16-kHz mono and decoded greedily with the English transcription prompt and at most 128 new tokens. The frozen deterministic aviation ITN baseline covers fixed call-sign prefixes, runways, headings, flight levels, decimal values, units, squawk codes, and selected aviation tokens. Evaluation-derived near-miss correction and sentence lookup are excluded from this general rule-based baseline. Spoken-prefix mappings require two unambiguous training groups; competing mappings require 10 groups and winner share 0.8. The single-adapter fusion exactly represents $0.68\Delta W_{0725} + 0.32\Delta W_{0820}$ as a rank-at-most-16 LoRA adapter. It doubles the stored adapter width relative to either endpoint but still uses one base-model forward pass and greedy decoding. Paired tests quantify test-example uncertainty for fixed checkpoints and do not estimate training-run variability.

Exact low-rank representation of adapter interpolation. For each adapted projection, let the two rank-8 endpoint updates be $\Delta W_A = B_A A_A$ and $\Delta W_B = B_B A_B$, with the common LoRA scaling absorbed into the B factors. Their weighted sum is represented exactly as

$$\begin{aligned}\Delta W_{\text{mix}} &= (1 - \alpha)\Delta W_A + \alpha\Delta W_B \\ &= \begin{bmatrix} (1 - \alpha)B_A & \alpha B_B \end{bmatrix} \begin{bmatrix} A_A \\ A_B \end{bmatrix}, \quad \alpha = 0.32.\end{aligned}\tag{1}$$

The concatenated inner dimension is $8 + 8 = 16$, so the resulting single adapter has rank at most 16. This is an exact factorization of the interpolated weight updates, not an elementwise average of the LoRA A and B factors.

Adaptive-analysis disclosure. The original Auxiliary, Entity Weighting, and Matched LoRA comparisons follow the original fixed-test protocol. Call-sign weakness for the second endpoint (random seed 20260820) was visible on development data, and its aggregate test degradation subsequently motivated adapter fusion and reference-free output routing. The fusion coefficient and routing rule were then selected on development data; neither procedure used a per-example test reference. Because the same test set had already been consulted, Tables S1–S3 label these results post-hoc and do not treat their intervals as independent confirmatory evidence.

Table S1: Complete post-hoc comparison on 4,487 test recordings. Normalized exact match and Entity F1 are percentages; WER is a ratio. The two-model system loads and decodes two adapters before deterministic reference-free routing.

System	ASR exact	ASR WER	Entity F1	Legacy exact	Legacy WER	Constr. exact	Constr. WER
Primary Auxiliary (seed 20260725)	33.56	0.5255	67.76	28.91	0.6618	32.85	0.5857
Second endpoint (seed 20260820, step 4,800)	35.10	0.4727	56.99	31.36	0.5829	34.48	0.5399
Single-adapter fusion ($\alpha = 0.32$)	37.02	0.4980	65.29	32.23	0.6246	36.39	0.5568
Two-model routed ensemble	36.73	0.4849	67.57	31.36	0.6123	36.13	0.5343

Table S2: Entity-value error decomposition for the primary Auxiliary and the single-adapter fusion. The F1 loss is recall-driven and concentrated in call signs.

Measure	Auxiliary	Fusion	Change
Precision (%)	89.86	90.02	+0.16
Recall (%)	54.39	51.22	−3.17
Micro F1 (%)	67.76	65.29	−2.47
True positives	824	776	−48
False positives	93	86	−7
False negatives	691	739	+48
Callsign true positives	296	248	−48
Callsign F1 (%)	61.60	55.11	−6.49

Table S3: Exploratory paired changes for Fusion minus the primary Auxiliary. Intervals use 20,000 paired bootstrap resamples and condition on the adaptively selected fusion; they do not correct for hypothesis generation.

Metric	Change	95% paired interval
ASR Exact (points)	+3.45	[+2.67, +4.28]
ASR WER	-0.0275	[-0.0531, -0.0034]
Entity F1 (points)	-2.47	[-3.63, -1.35]
Legacy Exact (points)	+3.32	[+2.63, +4.01]
Legacy WER	-0.0372	[-0.0659, -0.0095]
Unified Exact (points)	+3.54	[+2.79, +4.32]
Unified WER	-0.0289	[-0.0568, -0.0008]

Case-study conventions. The source course uses parenthetical expressions such as (call sign) as semantic placeholders rather than literal words to be produced. The frozen canonical evaluator removes these non-spoken placeholders and normalizes case and punctuation. Table S4 therefore illustrates entity preservation, structural conversion, and conservative fallback rather than punctuation generation; the selected cases are not used as aggregate evidence.

Table S4: Representative test cases illustrating the two roles of shared aviation structure. Red marks an ASR substitution and blue marks preserved or restored content; boldface retains the distinction in grayscale. Parenthetical placeholders are non-spoken course metadata removed before scoring.

(a) Entity-aware training avoids a domain-entity ASR substitution

Spoken-form reference	follow the follow me car to gate five zero seven you are number one
Canonical reference	(call sign), follow the follow-me car to gate 507. You are No. 1.
Matched LoRA	follow the follow me car to get to eight five zero seven you are number one
Entity-auxiliary LoRA	follow the follow me car to gate five zero seven you are number one

(b) The constrained module maps spoken entities to canonical forms

Spoken-form ASR	air cruiser four tree zero four roger contact departure on one one niner decimal seven they are aware of your emergency
Unified output	DPC4304 roger contact departure on 119.7 they are aware of your emergency
Canonical reference	DPC4304, roger, contact departure on 119.7, they are aware of your emergency.

(c) Conservative fallback preserves unsupported text

Input	hold position for further notification	<i>unchanged</i>
Output	hold position for further notification	
Canonical reference	(call sign), hold position for further notification.	

Table S5: Prompt-reference-input restoration diagnostic on all 5,922 development recordings. The input is the prompt-derived spoken-form reference, not a human verbatim transcript. Normalized exact match is a percentage and WER is a ratio. This diagnostic isolates restoration from recognition errors and was not used as a test-set result.

Restoration method	Norm. exact (%) ↑	WER ↓	Exact regressions ↓
Fixed aviation ITN	83.38	0.1059	39
Schema-constrained module	94.93	0.0140	0

Table S6: Canonical results by course-derived scenario for the primary Auxiliary recognizer. Normalized exact match is a percentage and WER is a ratio.

Scenario	N	Legacy ITN		Constrained module	
		Exact	WER	Exact	WER
Word/phrase reading	1,368	45.18	1.4177	45.18	1.4127
Sentence reading	99	17.17	1.0997	17.17	1.0960
Single-turn practice	2,420	27.11	0.5352	29.88	0.5012
Multi-turn practice	600	1.00	0.8520	19.33	0.5960

Table S7: Frozen training-only weak labels. (a) Distribution of the 17,076 spans generated from 48,098 references; no development or test reference contributes a span. (b) Examples with half-open whitespace-token intervals. The matcher also stores character offsets, and overlapping decoder subwords receive the listed class.

(a) Span distribution				(b) Span-alignment examples			
Type	Count	Type	Count	Matched surface	Token inter- val	Class	Canonical audit value
Call sign	5,657	Distance	2,965	<i>zenith one zero six</i>	[0, 4)	call sign	zenith:106
Runway	2,633	Time	2,129	<i>one two one decimal five</i>	[3, 8)	frequency	121.5
Heading	1,649	Flight level	839	<i>heading zero niner zero</i>	[13, 17)	heading	090
Altitude (m)	409	Frequency	325	<i>flight level one zero zero</i>	[9, 14)	flight level	FL100
QNH	273	Passengers on board	100				
Speed	95	Squawk	2				

Table S8: Frozen 50-recording development diagnostic. The subset contains 40 entity-positive single-turn scenarios and 10 entity-positive multi-turn exercises selected by deterministic hashes without reading any model output. Exact and Entity F1 are percentages; WER is corpus-level.

System	Exact	WER	Entity F1
Whisper large-v3	2.00	0.6198	51.69
Whisper large-v3 ATC FT	0.00	0.4802	70.37
Matched LoRA	6.00	0.3414	73.87
Entity-token weighting	6.00	0.3568	76.11
Entity-auxiliary LoRA	8.00	0.3342	67.89

The 50-recording subset was frozen on development data and is not a replacement for the complete test evaluation. It exposes the same metric-dependent trade-off as the main controlled comparison: Auxiliary has the lowest WER and one additional exact sentence, whereas Entity Weighting has the highest entity-value F1. The qualitative distance example in the main paper is the sole Auxiliary-only exact case against all displayed comparators in this fixed subset. The selection protocol does not read test data (test_used=false).

Table S9: Per-entity diagnostic on the 1,131 reference-entity-positive test utterances. Training weak-span counts are juxtaposed with test support and frozen-system performance for description only; no per-type result was used for model selection, and the table does not imply a causal relationship between label frequency and F1. P, R, and F1 are percentages.

Entity type	Train weak	Test	Matched	Weighting	Auxiliary		
	spans	support	F1	F1	P	R	F1
Call sign	5,657	615	59.52	60.80	85.55	48.13	61.60
Distance	2,965	102	82.16	80.42	91.36	72.55	80.87
Runway	2,633	294	64.96	65.68	92.26	52.72	67.10
Time	2,129	195	74.69	76.88	93.80	62.05	74.69
Heading	1,649	201	67.52	69.43	92.86	51.74	66.45
Frequency	325	103	82.68	80.00	92.21	68.93	78.89
Altitude (m)	409	3	100.00	100.00	75.00	100.00	85.71
Squawk	2	2	0.00	0.00	0.00	0.00	0.00
Micro	–	1,515	67.11	67.88	89.86	54.39	67.76

The aggregate scores mask substantial type imbalance. Call signs provide 615 of the 1,515 scored reference values and remain recall-limited, whereas distance and frequency have higher F1. Altitude and squawk estimates have too little support for substantive interpretation. The training matcher also labels flight level, QNH, passenger-count, and speed spans, which are not separate output types in this parser-based test diagnostic.